

Final Assessment - Practical – Development/Programming Assessment

Marks Break Down:

Question 1: 15 Marks

Question 2: 15 Marks

Question 3: 15 Marks

Question 4: 15 Marks

GitHub Profile – End to End - Assignments : 40 Marks

Course: Fullstack AI – Batch – 11

Theme of Paper:

Comprehensive Stock Price Research Blueprint

Maximizing NumPy, Pandas, Seaborn, Scikit-Learn, TensorFlow & Keras Usage

Since you already have CSV datasets, your objective should be **research-grade quantitative analysis**, not merely forecasting tomorrow's price. Professional quantitative analysts typically spend far more effort understanding market behavior, engineering predictive factors, and validating hypotheses than training neural networks.

1. Research Mindset

Instead of asking:

"Can I predict tomorrow's price?"

Ask:

- What factors drive returns?
- What drives volatility?
- What predicts trend continuation?
- What predicts trend reversal?
- Which indicators contain information?
- How do market regimes affect performance?
- Can patterns generalize across securities?

Research Framework:

```
1 Data Validation
2   ↓
3 Market Structure Analysis
4   ↓
5 Feature Engineering
6   ↓
7 Statistical Research
8   ↓
9 Machine Learning
10  ↓
11 Deep Learning
12  ↓
13 Backtesting
14  ↓
15 Risk Analysis
16  ↓
17 Research Conclusions
```

2. Maximum Usage of NumPy

Numerical Research Engine

Use NumPy for:

Log Returns

```
1 returns = np.log(
2     df['Close']/df['Close'].shift(1)
3 )
```

Annualized Volatility

```
1 annual_volatility = (  
2     returns.std() * np.sqrt(252)  
3 )
```

Drawdown Calculations

```
1 cum_return = np.cumprod(  
2     1 + returns  
3 )  
4  
5 running_max = np.maximum.accumulate(  
6     cum_return  
7 )  
8  
9 drawdown = (  
10     cum_return -  
11     running_max  
12 ) / running_max
```

Monte Carlo Simulations

Research:

- Future portfolio paths
- Risk estimation

```
1 simulations = np.random.normal(  
2     returns.mean(),  
3     returns.std(),  
4     10000  
5 )
```

Correlation Matrix

```
1 np.corrcoef(  
2     stock_returns.T  
3 )
```

Useful in portfolio research.

3. Maximum Usage of Pandas

Pandas should become your primary research platform.

Time-Based Features

```
1 df['Year'] = df.index.year  
2 df['Month'] = df.index.month  
3 df['Quarter'] = df.index.quarter
```

```
4 df['Day'] = df.index.dayofweek
```

Research:

- January effect
- Day-of-week effect
- Month-end effect

Rolling Statistics

```
1 df['MA20'] = (  
2     df['Close']  
3     .rolling(20)  
4     .mean()  
5 )  
6  
7 df['STD20'] = (  
8     df['Close']  
9     .rolling(20)  
10    .std()  
11 )
```

Momentum Factors

```
1 df['Mom5'] = (  
2     df['Close'] /  
3     df['Close'].shift(5)  
4 )  
1 df['Mom20'] = (  
2     df['Close'] /  
3     df['Close'].shift(20)  
4 )
```

Volatility Factors

```
1 df['Vol20'] = (  
2     df['Return']  
3     .rolling(20)  
4     .std()  
5 )
```

Volume Factors

```
1 df['Volume_Ratio'] = (  
2     df['Volume'] /  
3     df['Volume']  
4     .rolling(20)  
5     .mean()  
6 )
```

Research:

- Unusual volume
- Institutional participation

4. Maximum Usage of Seaborn

Most analysts underuse visualization.

Return Distribution

```
1 sns.histplot(  
2     df['Return'],  
3     kde=True  
4 )
```

Research:

- Fat tails
- Outliers
- Skewness

Correlation Heatmaps

```
1 sns.heatmap(  
2     df.corr(numeric_only=True),  
3     annot=True  
4 )
```

Find:

- Redundant features
- Hidden relationships

airplots

```
1 sns.pairplot(  
2     df[  
3         ['Return',
```

```
4     'Volume',
5     'Volatility',
6     'RSI']
7 ]
8 )
```

Research:

- Non-linear behavior
- Clusters

Regime Visualization

```
1 sns.boxplot(
2     x='Regime',
3     y='Return',
4     data=df
5 )
```

Study:

- Bull regimes
- Bear regimes
- High volatility periods

5. Advanced Feature Engineering

This is where alpha often originates.

Trend Features

```
1 MA20
2 MA50
3 MA100
4 MA200
```

Volatility Features

```
1 5-Day Volatility
2 10-Day Volatility
3 20-Day Volatility
4 60-Day Volatility
```

Price Range Features

```
1 df['HL_Range'] = (  
2     df['High'] -  
3     df['Low']  
4 ) / df['Close']
```

Gap Features

```
1 df['Gap'] = (  
2     df['Open'] -  
3     df['Close'].shift(1)  
4 )
```

Research overnight behavior.

Technical Indicators

Create:

- RSI
- MACD
- ATR
- Bollinger Bands
- Stochastic Oscillator
- ADX
- OBV

Research whether each contributes predictive power.

6. Maximum Usage of Scikit-Learn

Start with classical ML before neural networks.

Predictive Targets

Classification

```
1 Tomorrow Up  
2 Tomorrow Down
```

Regression

```
1 Tomorrow Return  
2 Future 5-Day Return  
3 Future 20-Day Return
```

Models

Logistic Regression

Market baseline.

```
1 LogisticRegression()
```

Random Forest

```
1 RandomForestClassifier()
```

Useful for:

- Feature importance
- Nonlinear relationships

Gradient Boosting

```
1 GradientBoostingClassifier()
```

Extra Trees

```
1 ExtraTreesClassifier()
```

Excellent for ranking features.

Validation

Never use:

```
1 train_test_split()
```

Instead:

```
1 TimeSeriesSplit()
```

Because markets are time dependent.

Pipelines

```
1 Pipeline([
2     ('scaler', StandardScaler()),
3     ('model', RandomForestClassifier())
4 ])
```

Production-ready methodology.

7. Research with Clustering

Often overlooked.

K-Means

Identify:

- Bull markets
- Bear markets
- Sideways markets

```
1 KMeans(n_clusters=4)
```

PCA

Reduce dozens of indicators.

```
1 PCA(n_components=5)
```

Research:

- Hidden factors
- Market structure

8. Maximum Usage of TensorFlow & Keras

Only after strong classical baselines.

Dense Networks

```
1 Sequential([
2     Dense(128, activation='relu'),
3     Dense(64, activation='relu'),
4     Dense(1)
5 ])
```

Use engineered features.

LSTM Research

Best for sequences.

Input:

```
1 60 Days History
2   ↓
3 Next Day Return
1 LSTM(128)
```

Study:

- Trend persistence
- Volatility forecasting

GRU Networks

Often outperform LSTM.

1 GRU(128)

CNN for Financial Time Series

Detect recurring patterns.

1 Conv1D()

Research:

- Breakouts
 - Reversals
 - Volatility shocks
-

CNN + LSTM Hybrid

Many institutional researchers test:

1 CNN
2 ↓
3 LSTM
4 ↓
5 Dense

Attention Models

1 MultiHeadAttention

Research:

- Long-term dependencies
- Market regimes

9. Important Research Topics

Volatility Prediction

Often easier than predicting prices.

Target:

1 Future_Volatility_20D

Regime Detection

Research:

1 Bull Market
2 Bear Market
3 Correction
4 Recovery

Trend Classification

Classify:

1 Strong Uptrend
2 Weak Uptrend
3 Neutral
4 Weak Downtrend
5 Strong Downtrend

Anomaly Detection

Use:

1 IsolationForest

Detect:

- Crashes
- Extreme rallies
- News-driven events

10. Professional Backtesting

Prediction means nothing without testing.

Create signals:

```
1 if probability > 0.60:  
2     BUY
```

Evaluate:

Sharpe Ratio

```
1 Sharpe =  
2 returns.mean() /  
3 returns.std()
```

Sortino Ratio

Focuses on downside risk.

Maximum Drawdown

Measure worst loss.

CAGR

Annualized return.

Calmar Ratio

```
1 CAGR /  
2 MaxDrawdown
```

Institutional favorite.

11. High-Impact Research Projects

Project 1

Feature Importance Ranking

Question:

Which market factors contain the most predictive information?

Tools:

- Pandas
 - Scikit-Learn
 - Seaborn
-

Project 2

Volatility Forecasting

Models:

- Random Forest
- XGBoost
- LSTM

Compare.

Project 3

Market Regime Detection

Using:

- Returns
- Volume
- Volatility

Apply:

- | | |
|---|--------|
| 1 | KMeans |
| 2 | PCA |
-

Project 4

LSTM vs Random Forest vs Gradient Boosting

Compare:

- Accuracy
 - Sharpe Ratio
 - Drawdown
-

Project 5

Multi-Factor Stock Ranking System

Factors:

- Momentum
- Trend
- Volatility
- Relative Strength
- Volume

Rank securities weekly.

Professional Quant Research Tips

1. Feature Engineering Matters Most

Approximate research value:

1	Feature Engineering	50%
2	Data Quality	20%
3	Validation	15%
4	Model Selection	10%
5	Hyperparameter Tuning	5%

2. Predict Returns, Not Prices

Prices are non-stationary.

Returns are more research-friendly.

3. Use Walk-Forward Testing

1	TimeSeriesSplit
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Always.

4. Build Baselines First

Before deep learning compare against:

1	Yesterday Return
2	Moving Average
3	Linear Regression
4	Random Forest

5. Explain Every Model

Use:

1	Feature Importance
2	Permutation Importance
3	SHAP

6. Study Failures

The most valuable research often comes from understanding:

- 1 When does the model fail?
- 2 Why does it fail?
- 3 Which regime causes failure?

7. Create a Research Journal

For every experiment document:

- 1 Hypothesis
- 2 Dataset
- 3 Features
- 4 Model
- 5 Parameters
- 6 Metrics
- 7 Results
- 8 Conclusions
- 9 Next Experiment

This single habit separates professional quantitative research from casual stock prediction projects.

Ideal End-to-End Tech Stack

- 1 CSV Data
- 2 ↓
- 3 NumPy (mathematics)
- 4 ↓
- 5 Pandas (feature engineering)
- 6 ↓
- 7 Seaborn (EDA)
- 8 ↓
- 9 Scikit-Learn (baselines, clustering, explainability)
- 10 ↓
- 11 TensorFlow/Keras (LSTM, GRU, Transformers)
- 12 ↓
- 13 Backtesting
- 14 ↓
- 15 Risk Analysis
- 16 ↓
- 17 Research Report

For a serious portfolio project, I would prioritize **Pandas → Seaborn → Scikit-Learn → SHAP → TensorFlow/Keras**, as this sequence delivers the highest research value and mirrors how many quantitative research teams actually work.

=====Dataset:

Dataset 1. Adobe (ADBE) Stock Data: 1986 to December 2024

[http://github.com/ShahzadSarwar10/Full-Stack-AI-Bootcamp-B-11/tree/main/FinalAssessment/Adobe\(ADBE\)Stock](http://github.com/ShahzadSarwar10/Full-Stack-AI-Bootcamp-B-11/tree/main/FinalAssessment/Adobe(ADBE)Stock)

Dataset 2 Microsoft Stock Details - Updated Regularly

<https://github.com/ShahzadSarwar10/Full-Stack-AI-Bootcamp-B-11/tree/main/FinalAssessment/MicrosoftStock>

Dataset 3: Oracle stocks price from 2019 to 2024

<https://github.com/ShahzadSarwar10/Full-Stack-AI-Bootcamp-B-11/tree/main/FinalAssessment/OracleStock>

Dataset 4: Salesforce (CRM) Stock Data:

<https://github.com/ShahzadSarwar10/Full-Stack-AI-Bootcamp-B-11/tree/main/FinalAssessment/SalesforceStock>

Steps - TO Do: [Neat, Clean, well-documented code]

Loading in DataFrame, Perform “Descriptive Statistics” Max- Operations, Input data analysis based on seaborn, Performing core operation of SK Learn, Perform metric analysis

[10 Marks for exceptional creative deep data analytical skills demonstration. For standard implementation – these marks will NOT be given.]

Need demonstration of all skills, learn in this course. End to end. For standard implementation, no marks , will be given. AI generated code or copy pasted code, will get only zero marks.

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